Pharmaceutical Laboratory Quality Control & Testing Analytics

End-to-End Oracle SQL / PL-SQL + Power BI Analytics Portfolio Project

Executive Summary

Developed an end-to-end pharmaceutical laboratory Quality Control analytics solution using Oracle SQL, PL/SQL, SQL*Plus and Power BI. The project manages laboratory testing data, product specifications, analysts, instruments, OOS results, retesting and turnaround time, and converts these data into actionable QC insights through an interactive three-page Power BI dashboard.

Business Problem & Objective

The objective is to provide a structured analytical system for monitoring laboratory testing activity, identifying Out-of-Specification (OOS) results, investigating product and parameter risk, evaluating analyst workload, monitoring instrument utilization, and tracking testing turnaround time.

Technology Stack

Oracle SQL • SQL*Plus • PL/SQL • Analytical SQL Views • Window Functions • Microsoft Power BI • DAX • Power Query • CSV data bridge for Oracle 10g XE

Project Scale

Metric	Value
Laboratory test records	1,010
Pharmaceutical products	10
QC analysts	10
Laboratory instruments	12
Testing parameters	10

Database Architecture

LABORATORY_RESULTS acts as the central testing table and connects to product, specification, analyst and instrument master data through foreign keys.

Core Tables: PRODUCTS, SPECIFICATIONS, ANALYSTS, INSTRUMENTS and LABORATORY_RESULTS.

SQL & PL/SQL Development

Applied joins, GROUP BY aggregation, CASE expressions, subqueries, date analysis, ranking, analytical/window functions, OOS analysis, retest analysis, turnaround-time analysis, product risk

classification, analyst performance analysis and instrument performance analysis. PL/SQL was used to generate laboratory test records while applying business rules.

Analytical Views

VW_QC_KPI • VW_PRODUCT_QC • VW_PARAMETER_QC • VW_ANALYST_PERFORMANCE • VW_INSTRUMENT_PERFORMANCE • VW_RETEST_ANALYSIS • VW_PRIORITY_ANALYSIS • VW_QC_DASHBOARD_DATA

Data & Analytics Workflow

Laboratory QC Data → Oracle Database → SQL/PL-SQL Processing → Data Validation → Analytical Views → Power BI-ready Dataset → Power BI Dashboard → QC Insights & Risk Analysis

Power BI Dashboard

1. QC Executive Dashboard — executive KPIs, monthly testing trend, OOS by product, product testing volume, parameter volume and analyst workload.
2. QC Investigation & Risk Analysis — PASS/OOS distribution, OOS by parameter, priority distribution, instrument workload, retesting and product OOS risk.
3. Analyst & Operational Performance — analyst workload, analyst OOS performance, analyst TAT, instrument TAT and analyst performance summary.

Validated QC KPIs

KPI	Validated Value
Total Tests	1,010
Passed Tests	960
OOS Tests	50
Pass Rate	95.05%
OOS Rate	4.95%
Average TAT	27.40 hours

Key QC Insights

- The dataset contains 1,010 laboratory test records with 960 PASS and 50 OOS results.
- The overall PASS rate is 95.05%, while the OOS rate is 4.95%.
- OOS records were redistributed across the 10 products for balanced analytical testing; each product has 5 OOS records in the final dataset.
- Retesting is linked to the OOS workflow, supporting investigation-oriented analysis.
- Turnaround time is analyzed at both analyst and instrument level to identify operational differences.

Technical Challenges & Solutions

- Oracle 10g to Power BI connectivity: a CSV bridge was used from the Power BI-ready Oracle view because the available Oracle connector environment did not provide the required client support.
- OOS distribution validation: the initial generated OOS records were concentrated on one product; a SQL correction redistributed OOS records consistently across all products.
- SQL*Plus execution: longer SQL/PLSQL operations were organized as executable .sql scripts to avoid multiline paste and SP2-0734 issues.

Skills Demonstrated

Oracle SQL • SQL*Plus • PL/SQL • Relational data modeling • Joins • Aggregations • CASE expressions • Subqueries • Window functions • Ranking • Data validation • QC analytics • OOS analysis • Retest analysis • TAT analysis • Power BI • DAX • Power Query • Dashboard design • Pharmaceutical laboratory domain analysis

Conclusion

This project demonstrates an end-to-end analytical workflow from structured laboratory QC data through Oracle SQL/PL-SQL processing and analytical views to an interactive Power BI reporting layer. It combines data analytics skills with pharmaceutical laboratory and quality-control domain knowledge.

Portfolio Positioning

Recommended keywords: Pharmaceutical QC Analytics, Laboratory Data Analytics, OOS Monitoring, Oracle SQL, PL/SQL, Power BI, DAX, Quality Control, Laboratory Testing, Analyst Performance, Instrument Analytics, Retesting and Turnaround Time.